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PES2UG24CS073

5B - CSE

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #10: Student Club Event Ticketing & Budget Portal
Deliverable 1: Requirements Table

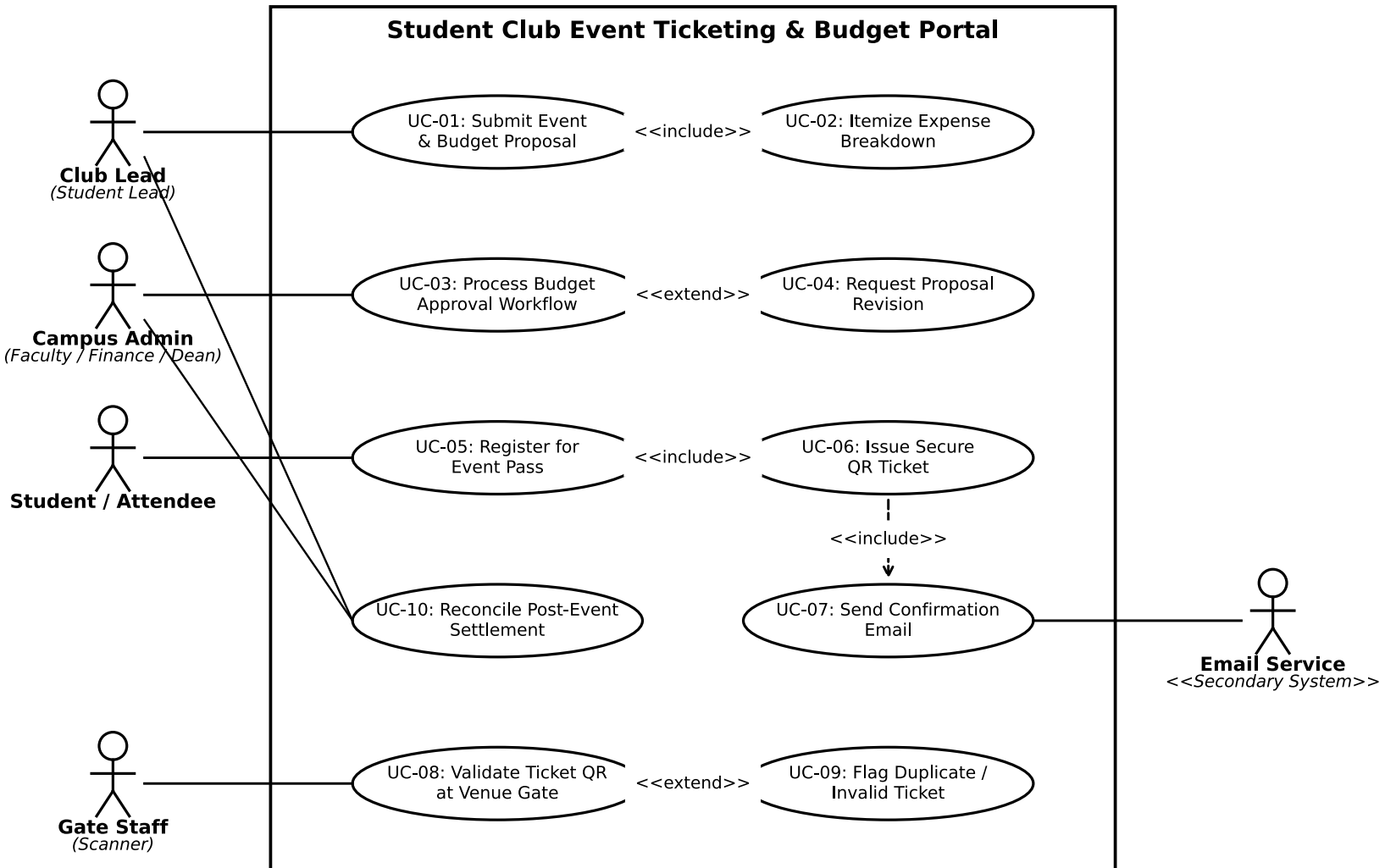
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall route event budget proposals through a multi-stage approval workflow (Faculty Coordinator -> Finance Office -> Dean).	High	Pass: Status transitions sequentially upon digital sign-off. Fail: Request reaches Dean without prior Finance Coordinator sign-off.	Enforces institutional governance, hierarchical financial checks, and audit compliance.
FR-002	Functional	The system shall allow club leads to draft event proposals, itemize budget line items (venue, logistics, catering), and upload supporting vendor quotations.	High	Pass: Proposal is saved with all line items and attachments in 'Under Review' state. Fail: Form submits with negative amounts or missing mandatory attachments.	Standardizes proposal submissions and captures structured cost projections.
FR-003	Functional	The system shall generate a cryptographically signed QR-code e-ticket and dispatch it via email upon confirmed student registration.	High	Pass: Registrant receives an email containing a unique, verifiable QR-code pass. Fail: Generates duplicate QR payload or fails to trigger email dispatch.	Automates pass distribution and prevents ticket tampering or duplication.

FR-004	Functional	The system shall provide an authentication scanner interface to validate attendee QR codes and record check-in timestamps in real time.	High	Pass: Successfully marks ticket as 'Checked-In' on first scan; immediately flags duplicate scans as invalid. Fail: Allows multiple entries using the same QR pass.	Prevents venue overcrowding, ticket re-entry, and pass-sharing fraud.
FR-005	Functional	The system shall generate post-event financial settlement reports comparing approved budget allocations against actual uploaded expenditure receipts.	Medium	Pass: Accurately computes variance and highlights discrepancies between allocations and bills. Fail: Variance calculations omit line items or miscalculate subtotals.	Facilitates fiscal audits and post-event fund reconciliation.
NFR-001	Performance & Security	The QR-code ticket scanner API shall validate entry authenticity in under 100 ms at venue check-in points.	High	Pass: Latency tests confirm response time < 100 ms under simulated peak load of 50 scans/sec. Fail: API response time exceeds 100 ms under standard load.	Eliminates attendee queue bottlenecks at entrance gates.
NFR-002	Security & Compliance	The system shall enforce Role-Based Access Control (RBAC) and encrypt financial transaction records at rest using AES-256 and in transit via TLS 1.3.	High	Pass: Unauthorized role access returns HTTP 403; all database audit logs capture actor ID, action, and timestamp. Fail: Unauthenticated users can view or alter financial requests.	Protects university financial data and administrative sign-offs from unauthorized access.

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Deliverable 2: UML Use-Case Diagram



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Deliverable 3: Use-Case Flow Document

Use Case ID & Name:	UC-01: Submit Event & Budget Proposal
Actors:	Club Lead (Primary), Email Notification Service (Secondary System)
Scope & Trigger:	Campus & Academic Operations | Club Lead initiates new proposal
Description:	Allows club leads to draft event details, itemize budget line items, attach vendor quotations, and submit the proposal into the multi-stage approval workflow.

1. Preconditions

- The Club Lead is authenticated with active student club administrative credentials.
- The university event submission window is active for the current academic session.

2. Postconditions

- The event proposal and budget records are saved in the system with status 'Pending Faculty Review'.
- Itemized line items and attached vendor quotation files are locked against further edits.
- An automated email notification with an authorization review link is dispatched to the Faculty Coordinator.

3. Main Success Scenario (MSS)

1. Club Lead navigates to the portal dashboard and selects 'Create New Event Proposal'.
2. System displays the event metadata form (Event Title, Category, Date, Venue, and Expected Headcount).
3. Club Lead fills in mandatory event metadata and selects 'Proceed to Budget Allocation'.
4. System invokes the budget breakdown module (<<include>> UC-02).
5. Club Lead inputs expense line items (Category, Description, Unit Rate, Quantity) and uploads vendor quotations.
6. System validates numerical inputs, calculates total requested budget, and displays an aggregated expense breakdown.
7. Club Lead reviews the complete proposal summary and clicks 'Submit for Approval'.
8. System validates all mandatory fields and confirms that total line-item costs evaluate to greater than zero.
9. System assigns a unique Tracking ID (e.g., PROP-2026-0042), sets status to 'Pending Faculty Review', and logs the action.
10. System triggers the Email Service to notify the Faculty Coordinator and renders a submission receipt to the Club Lead.
11. Use case terminates successfully.

4. Alternate Flows

4a. ***Invalid / Non-Numeric Budget Entries***

4a1. Club Lead inputs negative, zero, or non-numeric values for unit rates or quantities.
4a2. System highlights invalid fields with message: 'Unit cost and quantity must be positive numeric values.'
4a3. Club Lead inputs corrected numerical values; workflow resumes at Step 5.

5a. ***Missing Mandatory Vendor Quotation***

5a1. A budget line item exceeds the threshold requiring formal quotes, but no file is attached.
5a2. System flags line item: 'Official vendor quote attachment is required for this expense.'
5a3. Club Lead uploads valid quotation document; workflow resumes at Step 6.

7a. ***Save Proposal as Draft***

7a1. Club Lead selects 'Save as Draft' instead of submitting.
7a2. System stores current data with status 'Draft' and exits without initiating approver notifications.